MIL-T-82388(TD)
2 December 1969
SUPERSEDING
943-189A
4 August 1954

MILITARY SPECIFICATION

TRAINING AID; STAR PELORUS

This specification has been approved by the Naval Training Device Center, Department of the Navy

1. SCOPE

1.1 This specification covers the requirements for a hand-held, non-magnetic Star Pelorus training aid.

1.1.1 This item has been designated by the Naval Training Device Center as Device 1HH4 for Navy procurement.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bid or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated-plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-corner
PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation; Methods of
MIL-A-8625	Anodic Coatings; for Aluminum and Aluminum Alloys
MIL-L-10547	Liners, Case and Sheet, Overwrap, Water-vaporproof or Waterproof, Flexible
MIL-T-23991	Training Devices, Military; General Specification For

FSC 6910

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STANDARDS

Federal

FED-STD-406	Plastics, Method of Testing

Military

MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

Naval Training Device Center (NAVTRADEVCEN)

DL415200C60001	Training Aid, Star Pelorus

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Materials.- Materials for fabrication of the training aid shall be as specificed on drawing NAVTRADEVCEN DL415200C60001. Materials not specified on the drawing shall be as specified in 3.1.1 through 3.1.4.

3.1.1 Plastics.- Plastics shall be durable, resistant to cracking, brittleness, discoloration, deformation and shall be flame resistant. Plastic material when tested in accordance with Methods 2021, 2022 and 2023 of FED-STD-406 and judged as "nonburning by this test" or "self-extinguishing by this test" shall be considered NONFLAMABLE and may be used.

3.1.2 Metals.- Metals shall be aluminum (type 5052, 6061 or 6063) or be corrosion resistant in accordance with the corrosion-resistant metals requirements of MIL-T-23991.

3.1.2.1 Dissimilar metals.- Dissimilar metals, when used in intimate contact with each other, shall have provisions for protection against electrolysis and corrosion. The use of dissimilar metals in contact, which tend toward active electrolytic corrosion (particularly brass, copper or steel used in contact with aluminum or aluminum alloy) is prohibited.

3.1.3 Paint.- Paint for markings, dial graduations and indicators shall have a pigment composition equivalent in phosphorescence, durability, and color to Advance Process Supply Company's Light Green No. PP-30. Vehicles or solvents designed to be compatible with the pigments used, shall be used in sufficient quantities to cause the markings to be etched into the applied surfaces. The use of radioactive materials is prohibited.

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3.1.4 Cements.- Cements for joining plastics or for bonding plastics to other materials shall be those cements recommended by the manufacturers of the plastics. The cemented joints shall form a strong, homogeneous bond. The surface to be cemented shall be thoroughly softened by means of a solvent so that a soft layer or cushion is produced. The bond shall be effected by the uniting and hardening of the softened material. Bonding by capillary attraction shall require specific written approval of the procuring activity for each specific variance.

3.2 Design.- The design of the training aid is shown on the drawings of 3.1 and is a hand-held visual representation of a non-magnetic pelorus for training students in obtaining the azimuths of celestial bodies and relative bearings of landmarks. For purposes of taking readings at night, the dial graduations, lubber's line and indicators shall phosphoresce.

3.3 Reliability.- The training aid shall be constructed to withstand use in classrooms and in the field under adverse weather conditions for six hours per week for three years. The training aid shall not be adversely affected in any manner during storage, transportation and operation when under ambient conditions of temperature ranging from minus 5 to plus 125 degrees Fahrenheit and of relative humidity ranging to 100 percent. The training aid shall be assembled in such a manner that no fixed part or assembly shall become undesirably free or sluggish in operation and no movable part shall bind or seize. No degradation shall be caused in the mechanical performance stipulated for the training aid because of shipment, operation or storage.

3.4 Construction.- The training aid shall be constructed in accordance with the drawings of 3.1.

3.5 Details of components.- The training aid shall consist of the following major components and shall be in accordance with the drawings of 3.1.

(a) Alidade

(b) Relative bearing dial assembly

(c) Compass dial

(d) Sleeve

(e) Bolt and knurled nut.

3.6 Dimensions.- Dimensions and tolerances shall be in accordance with the drawings of 3.1. Dimensions and tolerances not shown on the drawings shall be selected to meet the requirements of 3.3.

3.7 Color and finish.- The training aid shall be lusterless black with white, light green, cream or ivory colored markings. All finished surfaces shall be free from toolmarks, burrs, scratches, sharp edges or other visual imperfections.

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3.7.1 Metals.- Aluminum shall be anodized black and be in accordance with MIL-A-8625. All other metals shall be primed in accordance with the recommendations of the manufacturer of the finish coat. The finish coat shall be baked-on black lusterless enamel.

3.8 Nameplates and product markings.- Nameplates and product markings will be specified by the procuring activity and shall be permanently affixed to the training aid for identification.

3.9 Government-furnished property.- Government-furnished property will be specified by the procuring activity.

3.10 Workmanship.- The training aid shall be constructed and finished with particular attention given to neatness, sturdiness, correctness of detail, freedom from sharp edges and burrs, ease of operation and clarity, sharpness and accuracy of graduations and lettering.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections.- Inspections to be performed are classified as follows:

(a) In-process inspection

(b) Quality conformance inspection.

4.2.1 In-process inspection.- The in-process inspection shall include such visual and mechanical examinations and testing of materials, subassemblies, parts and accessories (including purchased standard parts) during the manufacturing process as may be necessary to assure conformance to all the requirements of this specification.

4.2.2 Quality conformance inspection.- The quality conformance inspection shall consist of a preproduction inspection and an inspection of the production units.

4.2.2.1 A preproduction inspection shall be conducted on a preproductio unit and shall consist of all the examinations and tests specified in 4.3 and 4.4.

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4.2.2.2 Production units shall be inspected to the extent necessary to maintain quality control and conformance to the Government-approved preproduction unit.

4.3 Examinations.- Examinations shall be performed to verify conformance to the following requirements:

(a) Materials

(b) Construction

(c) Details of components

(d) Dimensions

(e) Color and finish

(f) Nameplates and product markings

(g) Workmanship.

4.4 Tests.- Each training aid shall be subjected to a functional test consisting of the following:

(a) Assemble training aid and check for ease of assembly

(b) With knurled nut loosened, rotate the alidade and compass disk independently 360°. Each part shall be free to rotate smoothly but with sufficient friction to assure alidade and disk maintains its respective setting if training aid is tilted 90° vertically

(c) With knurled nut tightened, rotate alidade 360°. The compass disk shall be locked sufficiently to rotate with the alidade.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be Level A or C as specified by the procuring activity.

5.1.1 Level A.- For Level A, each training aid shall be preserved and packaged for mechanical and physical protection in accordance with MIL-P-116, Method III.

5.1.2 Level C.- For Level C, preservation and packaging shall conform to the manufacturer's commercial practice.

5.2 Packing.- Packing shall be Level A or C as specified by the procuring activity.

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5.2.1 Level A.- Units packaged in accordance with 5.1.1 shall be packed in exterior containers conforming to PPP-B-621, Class 2, or PPP-B-601, Styles A or B with steel banding. The gross weight shall not exceed the weight limitations of the shipping container. Each exterior shipping container shall be lined with a waterproof moisture barrier sealed with adhesive or tape to effect water resistance in accordance with MIL-L-10547. A case liner shall not be required where the package(s) and package dunnage have been treated to resist water.

5.2.2 Level C.- Units packaged as specified in 5.1.2 shall be packed in containers which will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with common carrier rules and regulations as applicable to the mode of transportation.

5.3 Physical protection.- Each training aid shall be packaged disassembled in a container, conforming to PPP-B-636, Class 2, designed to provide a compact and nonshifting load. Each assembly or part shall be individually wrapped in a polyethylane or polyvinylidene chloride resin type of transparent film. Adequate cushioning, blocking and bracing shall be provided within shipping containers in a manner designed to prevent damage due to shock, vibration, impact or abrasion during shipment and storage.

5.3.1 Cushioning materials used shall be compatible with all finishes so that migrating solvents in cements, cushioning or components will not affect any part of the training aid.

5.4 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The star pelorus training aid is intended for use in training military and naval students in measuring azimuths of celestial bodies and relative bearings of landmarks.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification

(b) Selection of applicable levels of preservation, packaging and packing (see 5.1 and 5.2).

6.3 Definitions.- Except as indicated in 6.3.1 and 6.3.2, definitions will be furnished upon written request to the procuring activity.

6.3.1 Preproduction unit.- An item furnished under production contracts for the purpose of determining compliance with requirements of the specification already established and accepted. It shall be representative of production units though not necessarily made with production line jigs, fixtures and methods.

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6.3.2 Production unit.- A unit in its final mechanical form of final production design made by production tools, jigs, fixtures and methods.

CUSTODIAN:
Navy-TD

PREPARING ACTIVITY:
Navy-TD
Project No. 6910-N203

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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